

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Computer Science and Engineering
CO1 ASSESSMENT TOOL 2 – Agile Sprint Planning & User Story Workshop

Course Code:	CSA10 / CSA1063	Course Title:	Software Engineering
CO Assessed:	CO1	Assessment:	Assessment Tool 2 – Agile Sprint Planning & User Story Workshop
Weightage:	20%	Total Marks:	25
CO1 Statement:	<i>Apply advanced methodologies and agile project management techniques to manage software projects effectively</i>		
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Date:		Duration:	60 minutes

Code of Conduct

I K Y V Satyanarayana Reddy (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: *K Y V Satyanarayana Reddy*

Annexure A – Agile Sprint Planning & User Story Workshop

Business Intelligence Dashboard for Executive Decision Support

1. Identify the Project Vision and Stakeholders

- Define the project's vision, objectives, and identify all key stakeholders involved

Project Vision

The vision of the **Business Intelligence Dashboard for Executive Decision Support** is to develop an intelligent, centralized, and interactive dashboard that helps executives make accurate, timely, and data-driven decisions. The system integrates business data from multiple departments such as finance, sales, marketing, inventory, and human resources into a single platform. It provides real-time dashboards, Key Performance Indicators (KPIs), charts, graphs, and analytical reports that enable management to monitor organizational performance, identify trends, forecast future outcomes, and improve strategic planning. The proposed solution aims to eliminate manual reporting, improve operational efficiency, reduce decision-making time, and support sustainable business growth through effective business intelligence.

Project Objectives

- Develop a centralized Business Intelligence Dashboard.
- Integrate organizational data from multiple departments.

- Provide real-time business analytics and reporting.
- Improve executive decision-making using data visualization.
- Reduce manual report generation and reporting errors.
- Monitor Key Performance Indicators (KPIs).
- Improve organizational productivity and operational efficiency.
- Ensure secure access through role-based authentication.
- Support business forecasting and trend analysis.
- Enable faster strategic planning and business growth.

Key Stakeholders

Stakeholder	Role
Executives	Make strategic business decisions
Business Managers	Monitor departmental performance
Department Heads	Provide departmental business data
Data Analysts	Analyse and prepare business reports
Software Developers	Develop and maintain the dashboard
Database Administrators	Manage business databases
IT Administrators	Maintain servers and system infrastructure
Employees	Enter operational business data
Customers	Provide feedback and transactional information
Project Manager	Plan and coordinate project activities

2. Create Epics and User Stories

- Break down the requirements into Epics and formulate well-structured User Stories using the format:
As a <user>, I want <feature>, so that <benefit>

Epic 1: User Authentication

User Story 1

As an Executive, I want to log in securely so that I can access confidential business information safely.

User Story 2

As an Administrator, I want to manage user accounts so that only authorized users can access the dashboard.

Epic 2: Dashboard Management

User Story 3

As an Executive, I want to view real-time dashboards so that I can monitor business performance instantly.

User Story 4

As a Manager, I want to filter dashboard reports so that I can analyse departmental performance efficiently.

Epic 3: Report Generation

User Story 5

As an Executive, I want to generate business reports so that I can review organizational performance.

User Story 6

As a Manager, I want to export reports to PDF and Excel so that I can share them during meetings.

Epic 4: Data Analytics

User Story 7

As a Business Analyst, I want to analyse KPIs so that I can identify business trends.

User Story 8

As an Executive, I want predictive analytics so that I can make future business decisions confidently.

Epic 5: Notification System

User Story 9

As an Executive, I want to receive alerts when KPIs fall below target so that I can take corrective actions quickly.

3. Define Acceptance Criteria

- Write clear and measurable acceptance criteria for each selected user story using the Given–When–Then format or equivalent

User Story	Acceptance Criteria (Given–When–Then)
Login	Given a registered user, when valid credentials are entered, Then the dashboard should open successfully.
Dashboard	Given business data exists, When the dashboard loads, Then KPIs and charts should display correctly.
Report Generation	Given business data is available, When the user clicks Generate Report, Then the report should be created successfully.
Report Export	Given a report is generated, When Export is selected, Then the report should download in PDF or Excel format.
Notifications	Given KPI values cross predefined limits, when monitoring is active, then executives should receive automatic alerts.



4. Prioritize the Product Backlog

- Organize and prioritize the product backlog based on business value, customer needs, and project dependencies

Priority	Product Backlog Item	Business Value
High	User Login & Authentication	High
High	Dashboard Visualization	High
High	KPI Monitoring	High
High	Report Generation	High
Medium	Data Integration	Medium
Medium	Report Export	Medium
Medium	Notifications	Medium
Low	Predictive Analytics	Low
Low	Mobile Dashboard	Low
Low	AI Recommendation Engine	Low

5. Plan the Sprint Backlog

- Select high-priority user stories for the first sprint and divide them into actionable development tasks

Sprint 1 User Stories

- User Login
- Dashboard Home Page
- KPI Dashboard
- Basic Report Generation

Development Tasks

- Design login interface.
- Develop authentication module.
- Create dashboard layout.

- Connect dashboard with the database.
- Display KPIs using charts and graphs.
- Generate business reports.
- Perform unit testing.
- Fix identified bugs.
- Deploy Sprint 1 prototype.

6. Estimate Effort Using Story Points

- Assign story points to each selected user story using an appropriate estimation technique (e.g., Fibonacci sequence or Planning Poker) and justify your estimates.

User Story	Story Points	Justification
User Login	3	Simple authentication with moderate development effort.
Dashboard Visualization	8	Requires UI design, charts, and database integration.
KPI Monitoring	5	Needs business logic and data processing.
Report Generation	5	Requires report templates and export functionality.
Data Integration	13	Complex integration with multiple data sources.
Notifications	5	Requires background monitoring and alert generation.
Predictive Analytics	13	Involves AI/ML algorithms and advanced analytics.

Justification

The Fibonacci estimation technique is used because it helps estimate development complexity, uncertainty, and implementation effort. Larger story point values indicate higher complexity, greater

technical challenges, and increased development time, allowing the Agile team to plan sprints more effectively.

7. Present the Sprint Goal and Expected Deliverables

- Define a clear sprint goal and describe the expected sprint deliverables, including the features to be completed at the end of the sprint

Sprint Goal

The primary goal of Sprint 1 is to develop the foundational features of the Business Intelligence Dashboard that enable secure user access, display real-time business information, monitor Key Performance Indicators (KPIs), and generate basic business reports. At the end of the sprint, executives should be able to log into the system securely, access the dashboard, view business performance through interactive charts, and generate reports using real organizational data. Completing these core functionalities establishes a strong foundation for future sprints, where advanced features such as predictive analytics, AI-based recommendations, notifications, and mobile support will be implemented.

Expected Sprint Deliverables

- Secure user authentication system.
- User role management.
- Interactive dashboard interface.
- KPI monitoring module.
- Business report generation.
- Database connectivity.
- Basic data visualization using charts and graphs.
- Unit-tested software modules.
- Sprint review demonstration.
- Sprint documentation and project progress report.

Rubrics for Calculation

Criteria	Excellent (5 Marks)	Good (3 Marks)	Satisfactory (2 Marks)	Needs Improvement (1 Mark)
Project Vision & Stakeholder Identification	Project vision is clear, comprehensive, and aligned with the problem statement. All relevant stakeholders and their roles are correctly identified.	Vision is clear with minor omissions. Most stakeholders are correctly identified.	Vision is partially defined. Some stakeholders are missing or incorrectly identified.	Vision is unclear or incomplete. Stakeholders are poorly identified or missing.
Epics & User Stories	Epics are well-defined and user stories are complete, follow Agile format, and fully capture functional requirements.	Epics and user stories are mostly complete with minor formatting or requirement issues.	User stories are partially complete with limited Agile structure.	User stories are incomplete, unclear, or do not follow Agile practices.
Acceptance Criteria	Acceptance criteria are specific, measurable, testable, and aligned with all user stories.	Acceptance criteria are mostly clear and testable with minor gaps.	Acceptance criteria are partially defined and lack clarity or completeness.	Acceptance criteria are missing or not aligned with user stories.
Product Backlog Prioritization & Sprint Planning	Product backlog is logically prioritized, sprint backlog is realistic, and task breakdown is complete.	Backlog prioritization and sprint planning are mostly appropriate with minor issues.	Prioritization or sprint planning is partially complete with limited justification.	Backlog prioritization and sprint planning are poorly organized or incomplete.
Story Point Estimation, Sprint Goal & Presentation	Story point estimation is well justified, sprint goal is clear, deliverables are realistic, and presentation is professional and well organized.	Estimation and sprint goal are mostly appropriate with minor inconsistencies. Presentation is clear.	Estimation or sprint goal lacks justification. Presentation is adequate but incomplete.	Estimation is unrealistic or missing. Sprint goal, deliverables, and presentation are unclear or incomplete.

Criteria	Mark
Project Vision & Stakeholder Identification	/5
Epics & User Stories	/5
Acceptance Criteria	/5
Product Backlog Prioritization & Sprint Planning	/5
Story Point Estimation, Sprint Goal & Presentation	/5
TOTAL	/25